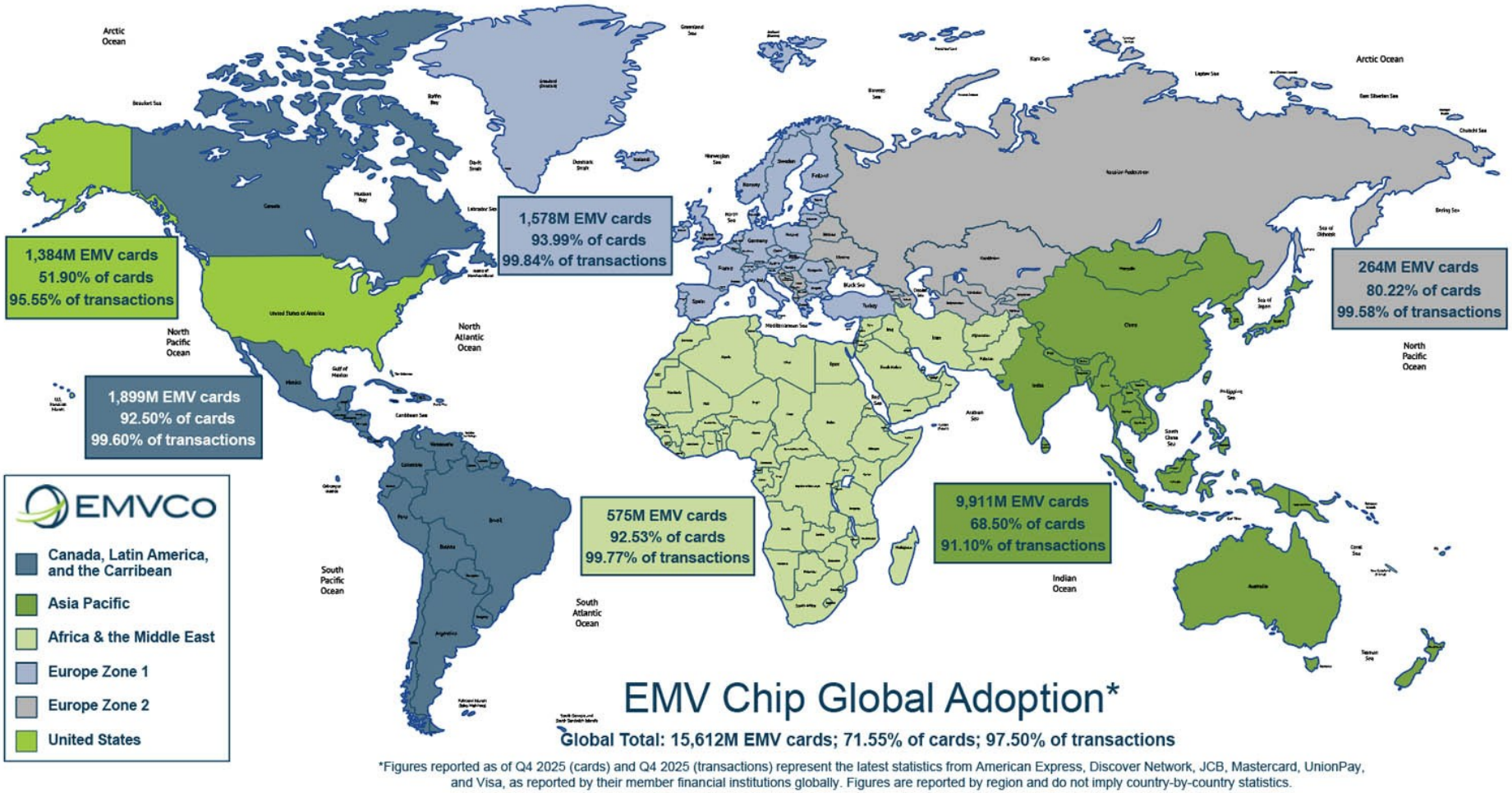
Weekly Reading

Look at the EMVCo website http://www.emvco.com/

- EMVCo sets specifications for the majority of smart payment cards (and mobile/payment tokens, etc.) used in electronic payments.
- Six main members
  - Mastercard
  - Visa
  - JCB
  - Union Pay
  - American Express
  - Discover

EMV Adoption



*Figures reported as of Q4 2025 (cards) and Q4 2025 (transactions) represent the latest statistics from American Express, Discover Network, JCB, Mastercard, UnionPay, and Visa, as reported by their member financial institutions globally. Figures are reported by region and do not imply country-by-country statistics.

EMV Encryption

Card security is discussed in EMV v4.4 Book 2 (Oct 2022)

a) Look at the specifications and identify the symmetric encryption algorithms are being recommended for use.

1) Double-length key triple DES (so 2 key 3DES)

2) AES (with 128, 192 or 256 bit keys)

Padding? Modes of Operation?

Pad with ’80’ then ‘00’s,

Use ECB or CBC modes (Contact)

Use CTR or CBC (Contactless)

b) Which algorithm would you use if you had a choice?

AES is the better algorithm for new developments…

CBC better than ECB